

tampering and environmental factors that traditional IT infrastructure may not encounter.

- **Scale and quantity:** IoT networks can consist of thousands or even millions of devices, far exceeding the scale of traditional IT devices and networks. Managing and securing this large number of endpoints poses a significant challenge.

Additionally, there are other challenges like life cycle management, supply chain vulnerabilities, and the lack of proper user interfaces.

Traditional security solutions are designed to protect IT systems such as enterprise networks, servers, and workstations and are not effective at safeguarding IoT devices due to these unique challenges.

How to ensure IoT security

Securing IoT requires a tailored approach, including measures such as device authentication, encryption, secure device management and regular updates.

- **Device security:** IoT devices need security at the hardware and software levels. We need to implement lightweight embedded agents that can keep the device's operating system and software up-to-date. In addition, they should offer runtime protection to monitor state, and respond when there are anomalies detected to identify and combat zero-day attacks.
- **Network security:** We need to implement a zero-trust strategy and segment the network to isolate

IoT devices from other devices, minimising the attack surface.

- **Firmware security:** In-depth examination of an IoT device's firmware should be conducted to detect potential vulnerabilities within its code.

IoT security solutions should also include auto device discovery and identification, secure firmware updates, and real-time monitoring and anomaly detection, using security information and event management systems to collect and analyse log data from IoT devices.

IoT security with LLMs

LLMs (large language models) are AI models based on artificial neural networks, trained using self-supervised learning and semi-supervised learning. As autoregressive language models, they work by taking an input text and repeatedly predicting the next token or word. They have the potential to significantly impact IoT security in various ways.

- **Automated threat detection:** LLMs can be used to develop natural language processing models to analyse logs, reports, and security alerts from IoT devices and networks. This can help in automated threat detection and response by identifying patterns and anomalies in textual data.
- **Automated patch management:** These models can automate security tasks like vulnerability scanning and improved patch management of IoT devices.

- **Security awareness and education:** LLMs can generate educational content, articles, and documentation about IoT security to raise awareness and understanding of IoT security best practices.
- **Comprehensive security policy development:** They can assist in creating and refining security policies and guidelines for IoT deployments.
- **Secure configuration templates:** LLMs can help generate secure templates for IoT devices and provide guidance on how to configure devices for better vulnerability management.

LLMs also have the potential to be used to develop new and more sophisticated IoT attacks that can have an adversarial impact on IoT security.

- **Adversarial attacks:** LLMs can be manipulated to generate deceptive content, including phishing emails, fake security advisories and malicious code that can lead to social engineering attacks against IoT users.
- **Automated exploitation:** LLMs can be used to automate malicious tasks, such as scanning for vulnerable IoT devices and exploiting them.
- **Privacy concerns:** Conversations between users and AI systems can disclose sensitive information about IoT devices and their vulnerabilities.

LLMs are proving to be highly effective at solving general-purpose tasks since they are trained on large data sets based on a broad generalist knowledge base. However, their ability to access specialised, private, and up-to-date knowledge is limited. To enhance LLM knowledge in a specific domain, fine-tuning them on domain-specific data sets is crucial.

It is important to use LLMs judiciously in IoT security to maximise their positive impact and mitigate the challenges. Human expertise is essential to validate the recommendations and findings generated by AI models. This includes implementing safeguards to

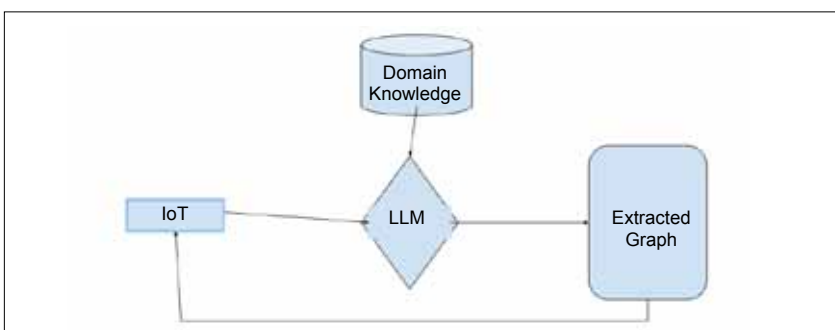


Figure 1: IoT security with LLMs